

Campus Entrance-Exit Monitoring System

A system that records students' time-in and time-out using student ID numbers for security and monitoring purposes.

Campus Problem Addressed

Lack of centralized digital records for monitoring student movement inside the campus. Manual logbooks are slow and difficult to audit for student attendance.

Suggested Core Features

- Student profile registration
- Student login via ID input (**ID Simulator**: A text field acting as a "Scanner" where students enter their ID number.)
- Time-in / time-out recording (View daily logs)
- View student logs/history (**Live Log**: A TableView that updates in real-time as students tap in/out.
- Generate simple reports: **Daily Attendance Report**: A feature to filter logs by date and export them.

Suggested Java Implementation

- **Java API**: Date & Time API, File Handling
- **JavaX**: Dashboard UI, TableView, input validation
- **JDBC**: CRUD operations for logs and student data

Possible Java APIs to Use

- `java.time.LocalDateTime`
- `java.io` (optional logging backup)
- JavaFX Forms + TableView

Database

Tables: Students, Attendance_Logs.

Expected Output

A desktop-based system that accurately tracks student campus presence.